Quantum Luthierie:
Translating Instrument Manufacturing Knowledge into Tangible Interfaces for Quantum-Computing Aided Musical Expression

Research Proposal

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1. Introduction

Quantum Computer Music (QCM) has emerged as a rapidly growing interdisciplinary field at the intersection of quantum computing and creative practice. The publication of two foundational volumes—Miranda's edited collection Quantum Computer Music: Foundations, Methods and Advanced Concepts [1] and the more recent Advances in Quantum Computer Music [8]—signals a maturing research community that now includes dedicated international symposia (ISQCMC), workshops at major venues such as IRCAM and SIGGRAPH, and an expanding body of peer-reviewed literature.

Within QCM, Quantum-computing Aided Composition (QAC), a concept introduced by Hamido [2][3], has demonstrated the potential of quantum circuit outputs for algorithmic music generation. The QAC Toolkit—a Max/MSP software package that allows musicians and artists to build, run, and simulate quantum circuits—has become the most widely adopted music-oriented framework for quantum computing [2]. Complementary work on OSC-Qasm [4] has further enabled interfacing between music software environments and quantum backends, while the Quantum Synth [5] and recent experiments in quantum-aided algorithmic composition across rhythm, timbre, harmony, and space [6] have expanded the palette of QAC techniques. The Intro to Quantum Harmony project [7] has specifically explored chord structures in superposition.

Throughout this body of work, Hamido has consistently articulated a deeper conceptual commitment: an interest in *the expressibility of electronics in interaction with live human performers* and a sustained inquiry into *the material means of artistic expression* [3]. This emphasis on materiality—on how the physical substrate of expression shapes the creative act—is a recurring thread. Yet the existing QAC ecosystem has primarily addressed this concern at the software level. The *physical, tangible dimension* of this material inquiry—how quantum-computing-based music generation connects to the bodies of performers, to the physical objects they manipulate, and to the centuries of embodied knowledge accumulated in the craft of instrument making—remains largely unexplored.

This proposal investigates how embodied knowledge from traditional instrument manufacturing can inform the design of tangible interfaces for QAC, bridging the gap between quantum-computing-based music generation and physical musical expression. The proposed research, titled Quantum Luthierie, draws on the researcher's dual background as both a trained multi-instrumentalist and an instrument maker with hands-on experience in saxophone key mechanism assembly, wind instrument quality control, and electric guitar setup.

2. Problem Statement

The QAC Toolkit currently operates through a Max/MSP patch-based interface, designed primarily for musicians with prior programming experience. For performers without a background in quantum physics, the mapping between quantum circuit outputs and musical parameters remains opaque—a black box whose internal logic does not afford intuitive physical interaction. This is not a limitation of the QAC framework itself, but rather a design opportunity: how can the QAC parameter space be made accessible through physical, tangible interaction?

This question intersects with two active concerns in the NIME community. First, there is growing interest in *tangible interfaces* that make complex computational processes physically graspable [9][10]. Second, the NIME community has increasingly engaged with questions of *sustainability and longevity* in digital instrument design [11]—how to build instruments that are repairable, adaptable over time, and capable of supporting a performer's evolving musical development.

Traditional instrument manufacturing contains centuries of accumulated knowledge about precisely these concerns. A saxophone key mechanism encodes specific relationships between physical parameters (key height, pad hardness, spring tension) and playability characteristics. This knowledge—what Polanyi [12] termed tacit knowledge and Schön [13] described as knowledge-in-action—represents a deep understanding of how millimeter-level physical design decisions shape musical expression. Yet this craft knowledge has never been systematically formalized as design principles for digital musical interfaces.

The gap is clear: no existing work has attempted to translate instrument manufacturing knowledge into design principles for quantum-computing music interfaces, nor has any QAC interface incorporated progressive disclosure mechanisms that adapt to a player's developing skill.

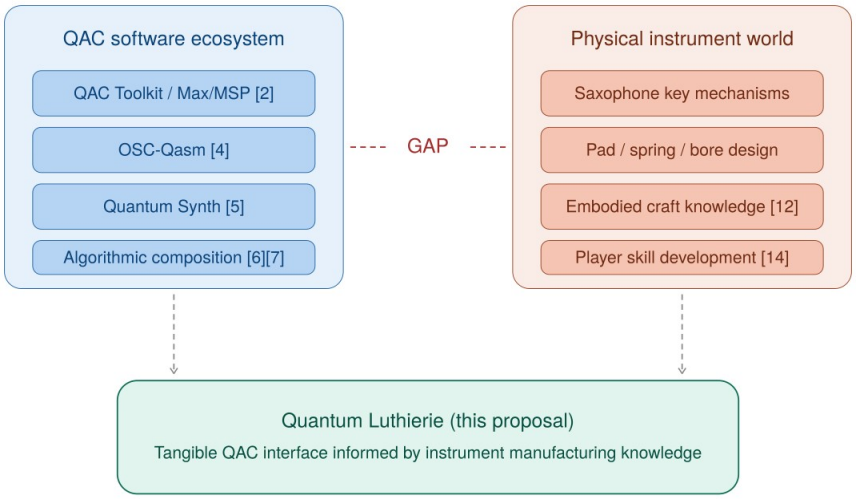

Figure 1. The research gap between QAC's software ecosystem and the physical instrument world.

3. Literature Review

3.1 Quantum Computer Music and QAC

Hamido's doctoral dissertation Adventures in Quantumland [3] provided the first systematic exploration of quantum computing's implications for music composition and performance. The subsequent QAC chapter [2] formalized the framework and introduced The QAC Toolkit. OSC-Qasm [4] enabled music software to interface with quantum backends via OSC, while the Quantum Synth [5] demonstrated real-time synthesis. Dobrian and Hamido [6] expanded QAC to rhythm, timbre, harmony, and space, and the Quantum Harmony project [7] explored chords in superposition. The Advances volume [8], including Mannone and Hamido's chapter on quantum memory, broadens theoretical scope. However, all work operates at the software level. Physical interface design for QAC has not been addressed.

3.2 Tangible Interfaces for Musical Expression

The Reactable [9] demonstrated that complex synthesis parameters could be made physically manipulable. McPherson and Kim [10] showed how embedded sensors could extend an instrument's expressive range. Masu et al. [11] raised sustainability concerns in NIME. Despite this body of work, no tangible interface has been designed specifically for quantum-computing music systems.

3.3 Tacit Knowledge, Craft, and Design

Polanyi's tacit knowledge concept [12] and Schön's reflective practitioner framework [13] provide theoretical foundations for understanding embodied craft expertise. No previous research has attempted to codify instrument manufacturing knowledge as design heuristics for digital musical interfaces.

3.4 Skill Acquisition and Progressive Disclosure

Dreyfus and Dreyfus [14] proposed a five-stage skill acquisition model informing progressive disclosure—a design pattern where interfaces gradually reveal complexity [15]. In traditional manufacturing, this is practiced implicitly: student-grade saxophones have simpler mechanisms than professional-grade instruments. This principle has never been applied to quantum-computing music interfaces.

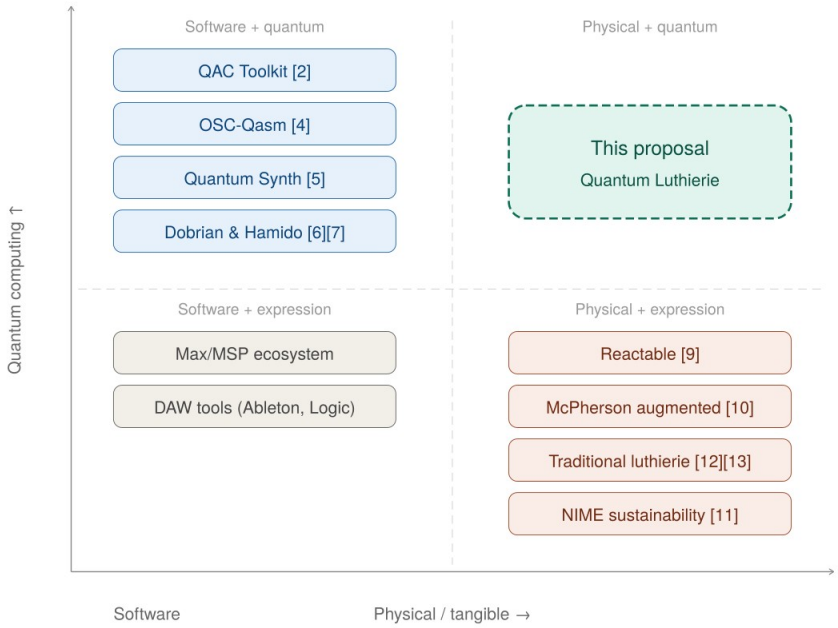

Figure 2. Literature landscape: the top-right quadrant (physical + quantum) is unoccupied.

4. Research Questions

RQ1 (Knowledge Extraction): How can tacit knowledge from traditional instrument manufacturing—particularly the mapping between physical parameters and playability in saxophone key mechanisms—be systematically documented and formalized into design heuristics for digital musical interfaces?

RQ2 (Interface Design): How can these heuristics inform the design of a tangible interface that enables performers to intuitively manipulate QAC parameters through physical interaction, building on the OSC-Qasm architecture [4], without requiring understanding of underlying quantum logic?

RQ3 (Progressive Disclosure): Can a progressive disclosure mechanism—grounded in Dreyfus's skill acquisition model [14] and the progressive disclosure design pattern [15]—where the interface initially offers limited, intuitive quantum parameter control and gradually exposes more complex parameter spaces as the player develops, meaningfully support musical expression and skill growth over time?

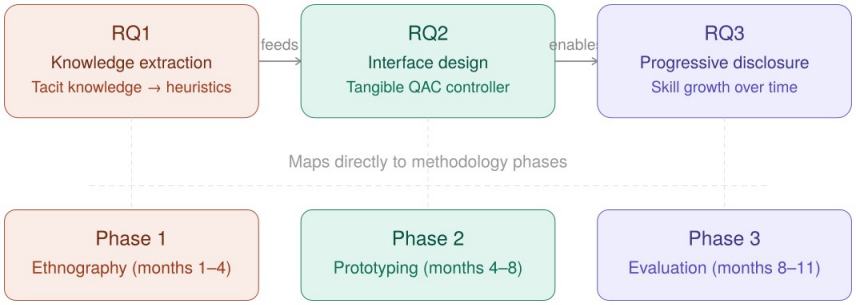

Figure 3. Research question dependency chain mapped to methodology phases.

5. Significance

This research is significant at multiple levels. Theoretically, it bridges QCM and NIME by extending quantum computing from a purely algorithmic tool into physical instrument design—a natural extension of the "material means of artistic expression" inquiry central to Hamido's work [3].

Practically, it creates an accessible entry point for performers without quantum physics backgrounds to engage with QAC-based creative practice. Methodologically, it demonstrates how traditional craft knowledge can serve as a design resource for emerging technology interfaces—generalizable beyond QAC.

For the xQAP group, the tangible controller will communicate with the QAC Toolkit via OSC protocol, building on OSC-Qasm [4]. Hardware schematics will be open-sourced on GitHub, and the Max/MSP integration patch will be a companion module for the QAC Toolkit package.

6. Methodology

The proposed research follows a practice-based research-through-design methodology in three phases over eleven months.

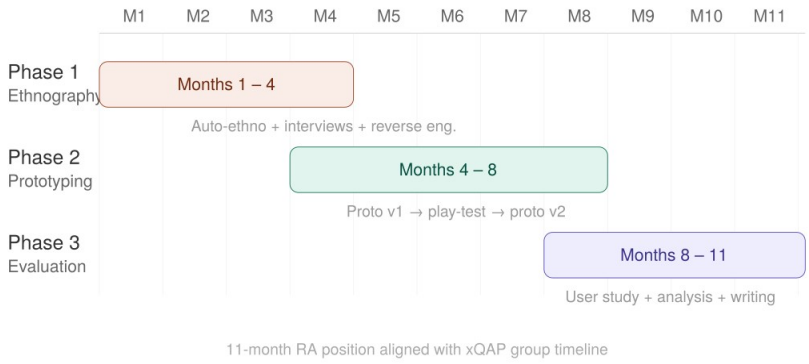

Figure 4. Eleven-month research timeline across three phases.

Phase 1: Ethnographic Documentation of Luthierie Knowledge (Months 1–4)

This phase adopts auto-ethnography as a primary method. Data sources: (a) *auto-ethnographic documentation* based on the researcher's manufacturing experience in Taizhou, Jiangsu; (b) *semi-structured interviews* with 3–5 craftspeople; and (c) *reverse-engineering analysis* of student-grade vs. professional-grade saxophones. Output: a codified set of Luthierie-to-Interface Design Heuristics.

Phase 2: Tangible QAC Interface Design and Prototyping (Months 4–8)

Hardware: Arduino/ESP32, pressure sensors, rotary encoders with adjustable damping, accelerometers, 3D-printed enclosures. Software: Max/MSP patch integrating QAC Toolkit via OSC protocol [4].

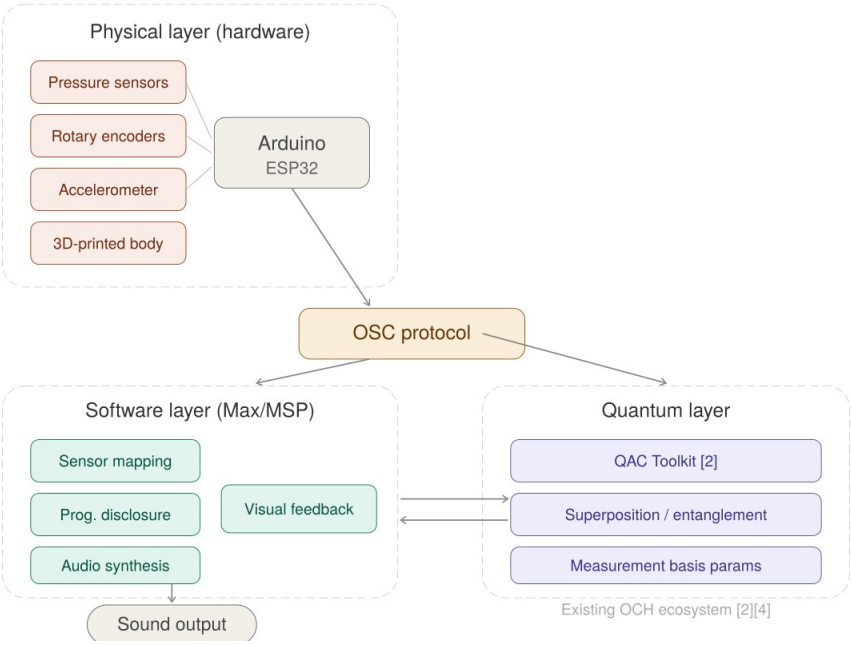

Figure 5. Tangible QAC controller system architecture.

The design translates quantum parameters (superposition probability, entanglement coupling, measurement basis) into physical interaction dimensions, guided by Phase 1 heuristics. A progressive disclosure mechanism implements three interface modes:

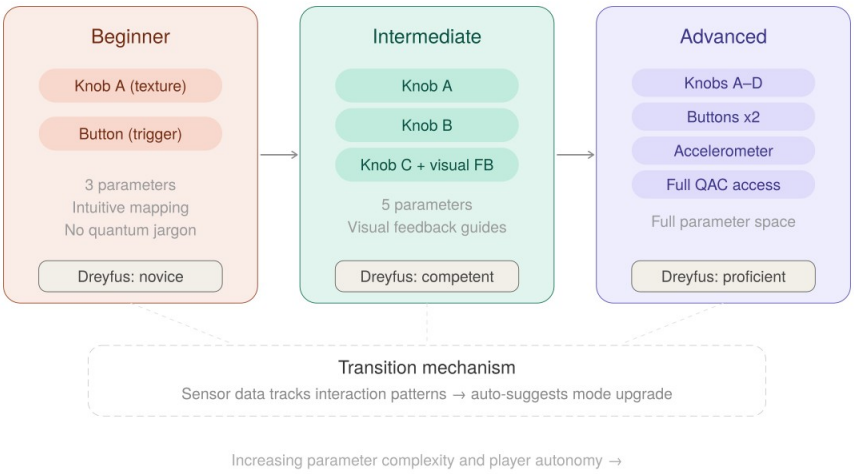

Figure 6. Progressive disclosure: three interface modes mapped to Dreyfus's skill stages.

Mode transitions are informed by sensor data tracking interaction patterns. At least two prototype iterations will be conducted based on play-testing feedback.

Phase 3: Practice-Based Evaluation (Months 8–11)

The researcher will use the prototype for a four-week creative practice period. Then 4–6 musicians will participate in a 4–6 week longitudinal study with within-subjects blind comparison: quantum parameters vs. pseudo-random classical generation. Data: performance recordings, sensor logs, interviews, and SUS.

7. Evaluation Metrics

Five metrics span quantitative and qualitative dimensions:

Interface Usability: SUS and NASA-TLX, targeting SUS > 68.

Parameter Space Utilization: Sensor/MIDI data analysis of parameter range exploration.

Perceived Expressiveness: Thematic analysis of interviews capturing creative freedom and musical agency.

Progressive Disclosure Effectiveness: Tracking natural migration from Beginner to Advanced modes.

Heuristic Validity: Expert review by 2–3 NIME-affiliated researchers.

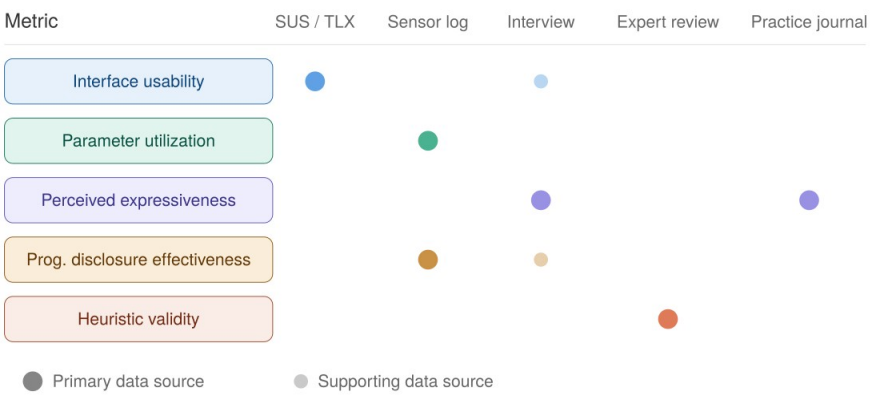

Figure 7. Mixed-methods evaluation framework: metrics mapped to data sources.

8. Expected Contributions

First, the Luthierie-to-Interface Design Heuristics—the first codification of instrument manufacturing tacit knowledge as design principles for digital musical interfaces, reusable beyond QAC.

Second, an open-source tangible QAC controller—hardware schematics, fabrication files, and Max/MSP integration—compatible with the QAC Toolkit and OSC-Qasm ecosystem [4].

Third, a progressive disclosure framework for quantum musical interfaces, contributing to NIME discourse on sustained musical growth.

Fourth, publications targeting NIME 2027 and ISQCMC 2027, directly aligned with xQAP's primary venues.

9. References

[1] Miranda, E. R. (Ed.). (2022). Quantum Computer Music: Foundations, Methods and Advanced Concepts. Springer.

[2] Hamido, O. C. (2022). QAC: Quantum-computing Aided Composition. In Miranda (Ed.), Quantum Computer Music (pp. 159–195). Springer.

[3] Hamido, O. C. (2021). Adventures in Quantumland. Doctoral dissertation, UC Irvine. eScholarship.

[4] Hamido, O. C., & Itaboraí, P. V. (2023). OSC-Qasm: Interfacing Music Software with Quantum Computing. EvoMUSART 2023, LNCS 13988 (pp. 372–382). Springer.

[5] Hamido, O. C., Cirillo, G. A., & Giusto, E. (2020). Quantum Synth. Audio Mostly Conference (pp. 265–268). ACM.

[6] Dobrian, C., & Hamido, O. C. (2025). Techniques for Quantum-Computing-Aided Algorithmic Composition. arXiv:2505.20565.

[7] Dobrian, C., & Hamido, O. C. (2024). Intro to Quantum Harmony: Chords in Superposition. MCM 2024, LNCS 14639 (pp. 147–158). Springer.

[8] Miranda, E. R. (Ed.). (2025). Advances in Quantum Computer Music. World Scientific. (Incl. Mannone & Hamido, Quantum Memory.)

[9] Jordà, S. et al. (2007). The reacTable. TEI ’07 (pp. 139–146). ACM.

[10] McPherson, A., & Kim, Y. (2012). The Problem of the Second Performer. Computer Music Journal, 36(4), 10–27.

[11] Masu, R. et al. (2021). NIME and the Environment. Proc. NIME 2021.

[12] Polanyi, M. (1966). The Tacit Dimension. University of Chicago Press.

[13] Schön, D. A. (1983). The Reflective Practitioner. Basic Books.

[14] Dreyfus, H. L., & Dreyfus, S. E. (1986). Mind over Machine. Free Press.

[15] Nielsen, J. (2006). Progressive Disclosure. Nielsen Norman Group.